Walk calibration of BTL modules in testbeam:

Via Synchronization of MCP scope data with FEBd

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Discussion draft; not approved for public release.

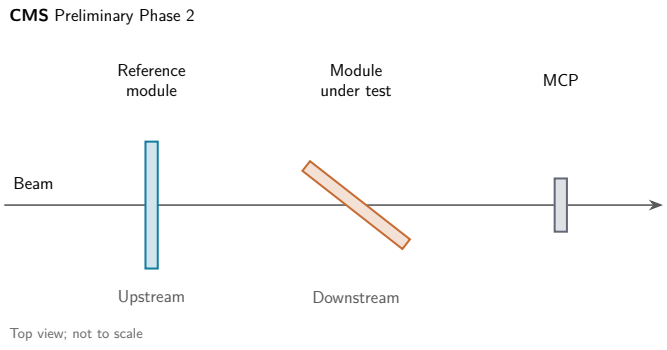
Abstract

Several analyses of the 2025 CMS barrel timing layer (BTL) test-beam data benefit from a reliable particle-arrival time reference. A Hamamatsu microchannel-plate photomultiplier (MCP-PMT), digitized by a digital oscilloscope, provides an external reference for a subset of the data. Its limited acceptance and acquisition rate motivate transferring the timing calibration to a BTL reference module.

A trigger link through the FEBd readout correlates events in the two streams. Clock alignment and locally weighted smoothing correct their low-frequency relative drift. Quadratic functions of inverse charge then describe the time walk of the left and right channels of a bar.

The calibrated reference bar provides a time reference for BTL analyses. The time difference between a bar in the module under test and the reference bar, before and after walk correction, tests the calibration. This FEBd-based synchronization procedure is specific to the test-beam setup.

Parallel MCP and BTL streams are linked through FEBd



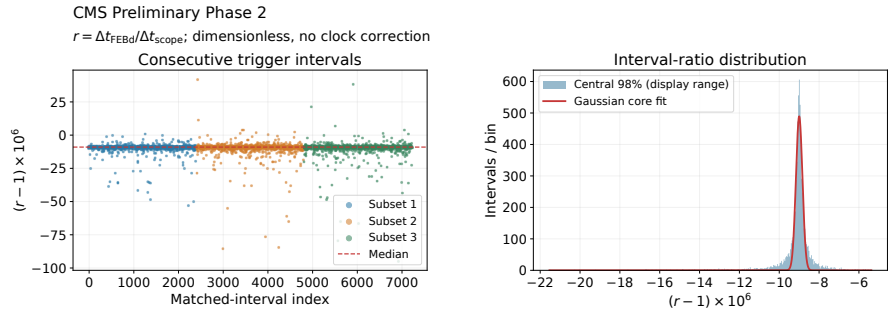
Two readout streams
The scope records the MCP-PMT pulse. FEBd records the BTL SiPM signals; the detector design is described in Ref. [3].

Synchronization link
The scope trigger also reaches a dedicated TOFHIR trigger channel in FEBd.
This link connects independently clocked records of the same particle.

September 2025 beam test, top view (not to scale). The reference module is perpendicular to the beam; the module under test is tilted. The MCP is downstream of both modules. This FEBd-based synchronization is specific to the test-beam setup, not the final HL-LHC synchronization scheme.

Events are correlated using the intervals between triggers

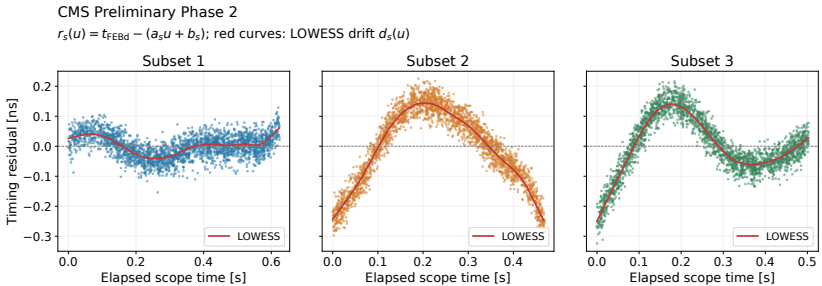
Match consecutive trigger-interval patterns in FEBd and the scope; retain unique event pairs [1].



Colors identify dataset subsets (segments). Δt is the interval between consecutive matched triggers within a subset. Scatter: all 7,203 intervals; histogram: central 98%. Axes show $(r-1)\times10^6$, a dimensionless quantity. Gaussian fit window: median $\pm1.5(1.4826\,\mathrm{MAD})$; its width measures ratio spread, not intrinsic timing noise. The next slide examines slow drift using time residuals after linear alignment. GitHub: trigger matching.

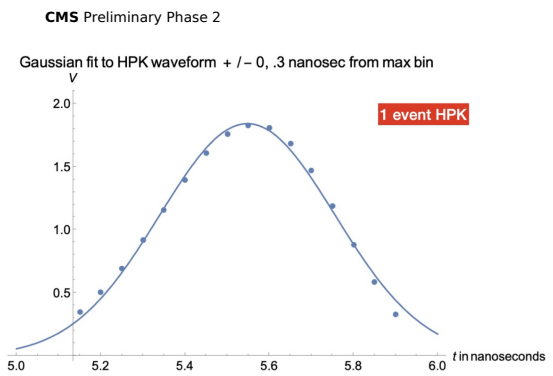
The matched clocks show low-frequency relative drift

Fit the relative clock rate in each subset, then model the remaining drift with LOWESS [7].



Same three subsets and colors as slide 4; points show all 7,206 matched events after separate linear fits, with shared ns limits. Red curves: robust locally weighted linear regression (LOWESS), using a 15% neighborhood and three robust iterations. Add the fitted drift to the line: $g_s(u) = a_s u + b_s + d_s(u)$. The calibration rejects subsets with pre-LOWESS scale $1.4826\,\mathrm{MAD} > 1$ ns; all three shown pass. W. S. Cleveland, JASA 74 (1979), 829–836 [7]. GitHub: clock reconstruction.

After synchronization, the MCP defines the calibration time



MCP characterization [2] and sampling checks [4] provide the waveform-timing context.
Within the scope waveform, define

$$\delta_{\mathrm{MCP}} = t_{\mathrm{peak}} - t_{\mathrm{trigger}}.$$

For SiPM channel i, use

$$\Delta t_i = t_i - g(u) - \delta_{\mathrm{MCP}}.$$

The charge-dependent part of this difference is the walk [5].

The waveform and fitted template determine the MCP pulse time. u is the scope-trigger timestamp. A fixed cable/geometry offset is irrelevant to the shape; clock-rate conversion over the short waveform interval is neglected.

Measure the reference-module channels' time walk versus charge

Separate threshold settings before selection and fitting.

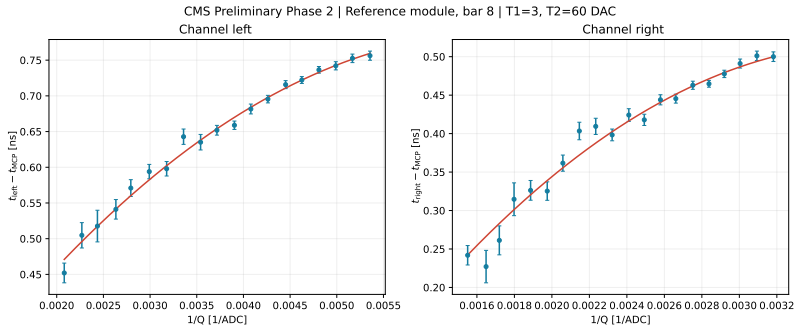
1. Select coincident events in the left and right channels of both the upstream reference bar and the downstream bar under test: four bar-end channels plus the dedicated trigger channel.
2. Within each setting, require clock quality and finite positive charge, ΔI and threshold-rise time. Keep charge above the spectrum's left half-maximum boundary.
3. Form 20 uniform $1/Q$ bins. Clip timing outliers at $5(1.4826\,\mathrm{MAD})$ from each bin's median; require at least five entries.
4. Fit the bin means with weights $1/\mathrm{SEM}^2$:

$$W_i(Q) = a_i + b_i/Q + c_i/Q^2.$$

Q is the measured ADC charge proxy, not energy in MeV. SEM denotes the standard error of the mean.

Reference module: bar 8; module under test: bar 3. Combine files only at identical scan settings. GitHub: selected reference sample.

The walk-corrected reference bar provides the time reference

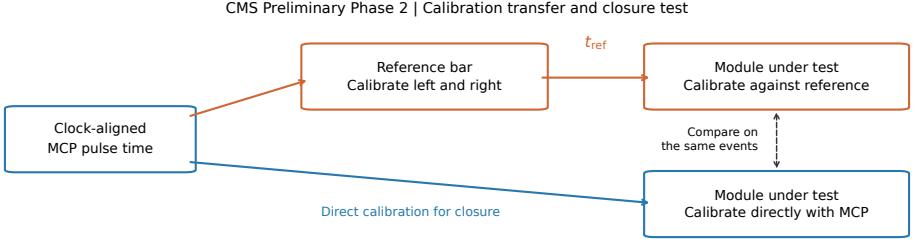


Correct each end: $t_i^{\mathrm{corr}} = t_i - W_i(Q_i)$. Use the bar average: $t_{\mathrm{ref}} = (t_L^{\mathrm{corr}} + t_R^{\mathrm{corr}})/2$.

Reference module, bar 8: left and right channels at $T_1 = 3$, $T_2 = 60$ (lowest scanned DAC setting). Points show profile means with SEM; curves are SEM-weighted quadratics in $1/Q$. t_{MCP} is the MCP pulse time mapped to the FEBd clock. The $+26$ ns offset preserves the shape. Fits are channel- and setting-specific; Ref. [6] discusses a slope-based alternative.

Transfer the MCP calibration to the module under test

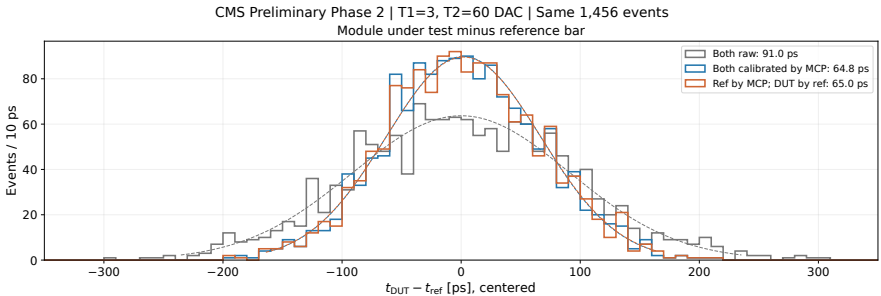
Purpose: use the calibrated reference bar when no MCP event time is available.



First form $t_{\mathrm{ref}} = \frac{1}{2}[(t_L - W_L^{\mathrm{MCP}}) + (t_R - W_R^{\mathrm{MCP}})]$. Then fit $t_j - t_{\mathrm{ref}}$ versus $1/Q_j$ for each end j of the module under test; subtract the resulting $W_j^{\mathrm{ref}}(Q_j)$.

The downstream transfer fit uses TOFHIR times and the saved reference-bar calibration, not the event's MCP time. For closure, also fit the downstream channels directly against MCP on the same selected sample. Both routes use 20 inverse-charge bins and SEM-weighted quadratics. One setting only: $T_1 = 3$, $T_2 = 60$ DAC. This tests transfer to one bar, not yet to every module or operating condition.

The transferred calibration agrees with direct MCP calibration



Both routes give a bar-pair width of about 65 ps, compared with 91.0 ps before correction.

Same 1456 events; common fitted charge domains, no extrapolation or threshold mixing. Bar times average the two corrected ends. Dashed curves: truncated Gaussian core fits within median $\pm 2.5(1.4826\,\mathrm{MAD})$; global median removed only for display. Width difference (via reference minus direct): $+0.24$ ps; 68% interval $[-0.30, 0.44]$ ps from 400 paired subset-bootstrap replicas. Cuts and calibrations fixed; no calibration or systematic uncertainty included. In-sample closure, not independent validation.

References

[1] GitHub repository, `https://github.com/De-Cristo/LecroyTranslator4CMSTimingDetector`.

[2] "Timing Performance of a Micro-Channel-Plate Photomultiplier Tube", `https://arxiv.org/pdf/1909.12604`

[3] CMS Collaboration, "A MIP Timing Detector for the CMS Phase-2 Upgrade". Technical Report CERN-LHCC-2019-003. CMS-TDR-020, CERN, Geneva, 2019 and "A High-Granularity Timing Detector for the ATLAS Phase-II Upgrade", CERN-LHCC-2020-007; ATLAS-TDR-031

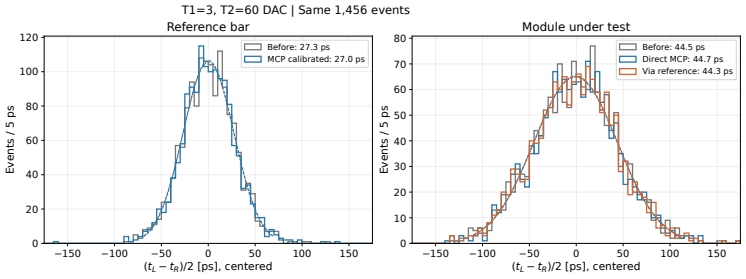
[4] S. White, Nov. 2025 BTL TB Analysis mtg presentation, `https://indico.cern.ch/event/1614521/contributions/6803723/attachments/3178820/5922530/5vs10vs20GSa_s.pdf`

[5] W. R. Leo, "Techniques For Nuclear and Particle Physics Experiments", p.326, Springer-Verlag Berlin Heidelberg.

[6] S. White and A. Boletti, "Amplitude Walk in Fast Timing: The Role of Dual Thresholds" 2026 JINST V21 P02001.

[7] W. S. Cleveland, "Robust Locally Weighted Regression and Regression Scatterplots", J. Amer. Statist. Assoc. **74** (1979) 829.

Backup: left-minus-right timing check



For $t_{\mathrm{bar}} = (t_L + t_R)/2$, estimate $\sigma_{\mathrm{bar}}^{(L-R)} = \sigma(t_L - t_R)/2$. Combine the bar estimates: $\sqrt{27.0^2 + 44.3^2} = 51.9$ ps, compared with the measured 65.0 ps bar-pair width.

The plotted variable is $(t_L - t_R)/2$: widths, fit curves and bins are rescaled by one half, with no refit. Same events as the closure test. Interpreting these widths as bar-average resolutions requires independent end fluctuations and negligible position spread; quadrature also assumes independent bars. Common timing fluctuations cancel in the half-difference. Correlations and position effects require further study to explain the 65 ps width.

Method for Walk Correction using Leading Edge Slope:

Derived from dual threshold feature of TOFHIR2C

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Abstract

Precision timing at the HL-LHC requires rapid, repeatable calibration as detector response evolves. With many interactions per bunch crossing, an external particle-arrival time reference is not readily available. The CMS barrel timing layer (BTL) therefore motivates calibration procedures based on internal measurements and a consistent parametrization across about 330 000 channels.

The September 2025 test beam exposed BTL sensors with TOFHIR2C readout to 180 GeV pions. Separate acquisitions of the same reference module provide dual-threshold slopes and MCP-referenced timing; they contain different events. Extending an earlier bench study, the mean slope versus charge defines an inverse-slope walk correction with one amplitude parameter, AWC, per SiPM and configuration.

Comparison with independent microchannel-plate (MCP) reference calibrations shows good agreement between the correction shapes. Internal charge and slope information provide candidate predictors of AWC, with the MCP results serving as benchmarks. Predictive precision and transfer to other operating conditions remain to be validated for reference-free calibration.

BTL and TOFHIR2C: Refs. [3, 7]; dual-threshold motivation and bench study: Ref. [5].

Charge-based calibration works, but its coefficients vary

The MCP reference note obtains good walk descriptions with a quadratic in inverse charge [6]:

$$W_i(Q) = a_i + b_i/Q + c_i/Q^2.$$

Bar	SiPM	a_i [ns]	b_i [ns ADC]	c_i [10^4 ns ADC^2]
7	Left	0.080 ± 0.067	238.3 ± 35.8	-2.16 ± 0.47
7	Right	-0.515 ± 0.090	422.6 ± 68.0	-5.58 ± 1.27
8	Left	0.122 ± 0.027	198.5 ± 13.1	-1.52 ± 0.16
8	Right	-0.229 ± 0.033	388.9 ± 26.2	-5.13 ± 0.51

With Q expressed as raw, uncalibrated ADC values, the shape coefficients vary substantially across these four SiPMs.

Original unweighted fits; 1σ errors assume independent profile means, fixed bins and selection. No systematics; coefficients strongly correlated. Q is not gain-calibrated; a_i includes an arbitrary offset. Coefficient variation does not measure calibration precision. Earlier experience: ALICE [1, 2].

TOFHIR2C measures the leading-edge slope with two thresholds

Larger pulses cross a fixed threshold earlier: the timestamp depends on amplitude [4]. Two-threshold timing has earlier implementations [8, 9]; TOFHIR2C provides both crossings [7, 5].

$$I_1 < I_2, \qquad t_1 < t_2, \qquad S = \frac{I_2 - I_1}{t_2 - t_1} \quad [\mu\mathrm{A/ns}].$$

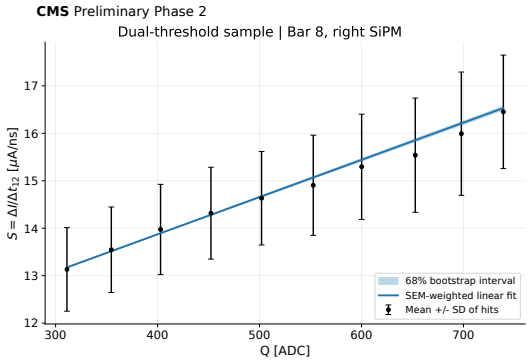
Locally, an approximately linear leading edge gives

$$\Delta t \simeq \frac{\Delta I}{S}.$$

Physical model: describe amplitude walk with an inverse slope, rather than an independent polynomial shape for every channel.

S is a finite-interval slope proxy, not the instantaneous derivative. The dual-threshold sample uses $\Delta I = 7.8125\,\mu\mathrm{A}$ from the implemented threshold conversion. The fitted AWC need not equal this threshold separation. GitHub: timing and threshold reconstruction [11].

Measure the charge-to-slope mapping using internal hits



Typical single-hit relative uncertainties: Direct slope: $\sim 15\%$; charge: $< 5\%$. Use the more precise Q to estimate the mean slope:

$$\overline{S}_i(Q) = c_{0,i} + c_{1,i} Q.$$

Ten means; weights $1/\mathrm{SEM}^2$; no MCP time.

Reference module, bar 8, right SiPM: 19 312 hits. Both SiPMs required; other module unconstrained. Physical crossings; peak-right charge cut. Points: means; bars: hit standard deviation (SD). Standard error of the mean: $\mathrm{SEM} = \mathrm{SD}/\sqrt{N}$, for N hits per bin. Band: conditional 68% interval, 1000 hit-bootstrap refits, fixed bins.

One amplitude describes walk once the slope is calibrated

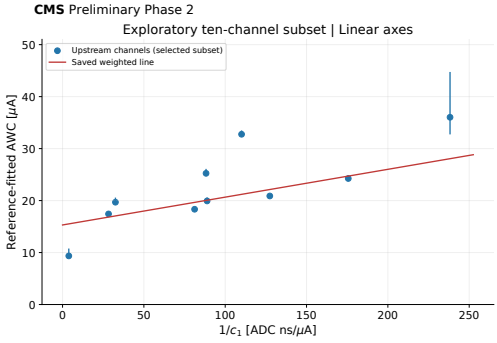
$$W(Q) = \frac{\mathrm{AWC}}{\overline{S}(Q)}, \qquad t_{\mathrm{corr}} = t_{\mathrm{measured}} - W(Q).$$

- Internal hits determine $\overline{S}(Q)$, the mean slope at a given charge, for each SiPM.
- One parameter, AWC (in μA), sets the walk magnitude.
- AWC is constant per SiPM and configuration; the correction varies with Q.
- Channel timing offsets are a separate calibration.

The remaining task is to predict AWC from internal observables. Use the MCP-calibrated channels to develop and test that prediction.

Apply the model within its calibrated charge range, with $\overline{S}(Q) > 0$. The slope fit alone does not fix AWC. The common functional form follows the dual-threshold model in Ref. [5].

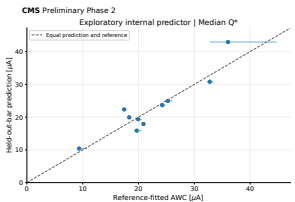
Candidate 1: AWC versus inverse slope coefficient



An exploratory trend: AWC versus $1/c_1$, the inverse slope-to-charge coefficient. Each point is one SiPM, not a charge bin. AWC is benchmarked against MCP; c_1 uses internal hits. The ten-SiPM trend motivates prediction, but the line is not a precision calibration ($\chi^2/\mathrm{ndf} = 538.6/8$). Bars: slope-only 16–84% bootstrap intervals; reference errors and correlations are omitted. More independent data are needed.

Candidate 2: a physics-motivated AWC predictor

Physics motivation: similar scintillation pulses and front-end shaping motivate testing a shared timing sensitivity to the same fractional charge change.



$$K \simeq -\left.\frac{dW}{d\ln Q}\right|_{Q_*}, \quad \mathrm{AWC}_{\mathrm{pred}} = \frac{K\overline{S}(Q_*)^2}{c_1 Q_*}.$$

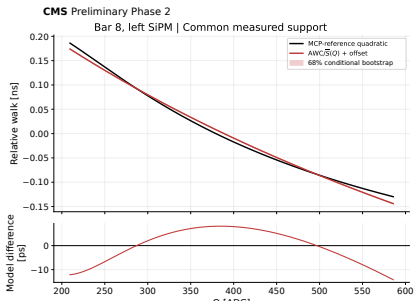
Required assumptions

1. $W = \mathrm{AWC}/\overline{S}$ holds locally, with constant AWC per SiPM.
2. The fitted $\overline{S}(Q) = c_0 + c_1 Q$ is reliable near Q_*.
3. Q_*, the selected-charge median per SiPM, samples comparable pulse regions.
4. Similar pulses and operating conditions allow a common K across SiPMs and bars.

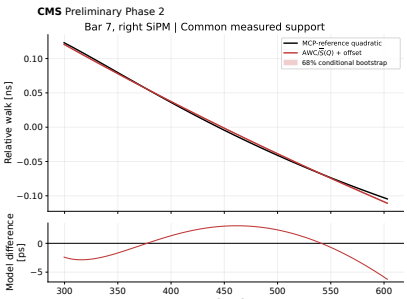
Leave-one-bar-out test: ten SiPMs, six bars; K learned from other bars' MCP AWC. RMS: 3.22 μA (14.1%). Dashed: equality. Bars: conditional reference-AWC intervals, not prediction intervals. Full uncertainties and correlations unvalidated. Transfer across thresholds, overvoltage or charge-integration settings untested; recalibration may be needed.

The single-amplitude model closely follows the MCP curves

Bar 8, left SiPM

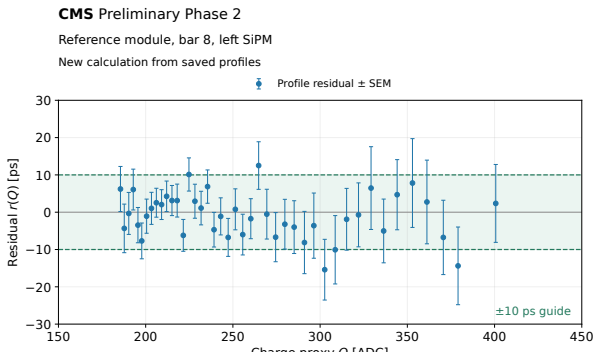


Bar 7, right SiPM



Reference module; independent samples. Black: MCP quadratic; red: dual-threshold model. AWC and offset minimize squared curve differences on equally weighted Q-grid points in common populated intervals, excluding gaps. Bar 8, left SiPM: 6.80 ps RMS mismatch. Lower panels: curve differences, not resolution. Bands: slope-fit statistics, MCP curve fixed. Clock alignment: LOWESS [10]. Shape benchmark, not AWC prediction.

The AWC correction describes the walk at the 10 ps scale



- **Y axis:** mean time minus MCP, the AWC correction and a constant offset.
- **43 charge bins:** within the slope calibration range, with 6.1 ps RMS of bin residuals.
- ± 10 **ps guide:** indicates the residual-walk scale, not an all-point bound.

$r(Q) = \langle t_{\mathrm{thr1}} - t_{\mathrm{MCP}} \rangle_Q - \mathrm{AWC}/\overline{S}(Q) - b$; b: fitted constant. New MCP-profile calculation [6]: saved $\overline{S}(Q)$, fixed $\mathrm{AWC} = 20\,\mu\mathrm{A}$. Bars: input profile-mean SEM, conditional on fixed correction and offset. Guide: not a confidence band. This is neither single-hit resolution nor AWC-prediction validation.

References

[1] "Performance of the ALICE Time-Of-Flight detector at the LHC", Eur. Phys. J. Plus (2013) 128: 44

[2] "PID performance of the ALICE-TOF detector at Run 2", N. Jacazio in Proceedings of LHCP2018, https://pos.sissa.it/321/232/.

[3] CMS Collaboration, "A MIP Timing Detector for the CMS Phase-2 Upgrade". Technical Report CERN-LHCC-2019-003. CMS-TDR-020, CERN, Geneva, 2019 and "A High-Granularity Timing Detector for the ATLAS Phase-II Upgrade", CERN-LHCC-2020-007; ATLAS-TDR-031

[4] W. R. Leo, "Techniques For Nuclear and Particle Physics Experiments", p.326, Springer-Verlag Berlin Heidelberg.

[5] S. White and A. Boletti, "Amplitude Walk in Fast Timing: The Role of Dual Thresholds" 2026 JINST V21 P02001.

[6] S. White and Licheng J. Zhang, "September 2025 BTL Testbeam Reference Bar Walk Calibration from Hamamatsu MCP-PMT ",cms note.

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[9] A.R. Frolov, T.V. Oslopova, Yu.N. Pestov, "Double threshold discriminator for timing measurements", Nuclear Instruments and Methods Volume 356, Issues 2/3 (1995) https://doi.org/10.1016/0168-9002(94)01239-3.

[10] W. S. Cleveland, "Robust Locally Weighted Regression and Regression Scatterplots", J. Amer. Statist. Assoc. **74** (1979) 829.

[11] GitHub repository, https://github.com/De-Cristo/LecroyTranslator4CMSTimingDetector.